

FUNCTIONAL ANALYSIS

(5 hours)

January 15, 2026

Please write on the front page your preferred time for the oral exam.

1. Let $(X, \|\cdot\|)$ be a normed space.

a) Show that a linear functional l on X is continuous if and only if its null-space $N_l = \{x \in X : l(x) = 0\}$ is closed in X .

b) Prove that if X is finite dimensional then every linear functional on X is continuous.

c) Let Y be a subspace of X and let $(U, \|\cdot\|_U)$ be a **finite dimensional** normed space. Show that every map $M \in \mathcal{L}(Y, U)$ extends to a bounded map $\tilde{M} \in \mathcal{L}(X, U)$ with the same norm.

2. a) Given a Hilbert space H , a closed subspace K and $x \in H$ prove that the vector $y = y(x) \in K$ is the (unique) closest element to x , that is,

$$\inf_{y \in K} \|x - y\| = \|x - y(x)\|,$$

if and only if $x - y(x) \in K^\perp$.

b) Find $y(x)$ when $H = L^2[-1, 1]$ and $K = \{y \in L^2[-1, 1] : y(t) = 0 \text{ a.e. on } [-1, 0]\}$.

c) Find $y(x)$ when $H = L^2[-1, 1]$ and $K = \{y \in L^2[-1, 1] : y(t) = y(-t) \text{ a.e.}\}$.

3. Let $(X, \|\cdot\|_X)$, $(Y, \|\cdot\|_Y)$ be Banach spaces and let $M \in \mathcal{L}(X, Y)$.

a) Assume that $R_M = M(X)$ contains a closed subspace Z of Y . Prove that there exists a constant $c > 0$ such that for every $z \in Z$ there exists $x \in X$ with $Mx = z$ and $\|x\|_X \leq c\|z\|_Y$.

b) Does the conclusion from a) hold for arbitrary subspaces $Z \subseteq R_M$? Justify your answer.

4. Let $C^2([0, 1])$ be the Banach space of twice continuously differentiable functions on $[0, 1]$ with the norm

$$\|f\|_2 = \sup_{t \in [0, 1]} |f''(t)| + \sup_{t \in [0, 1]} |f'(t)| + \sup_{t \in [0, 1]} |f(t)|.$$

Let $a, b, c \in C([0, 1])$ and assume that for every $g \in C([0, 1])$, the differential equation

$$af'' + bf' + cf = g$$

has a solution $f \in C^2([0, 1])$ (which is not necessarily unique). Show that there exists a constant $k > 0$ such that for every $g \in C([0, 1])$ there exists a solution $f \in C^2([0, 1])$ of the above equation with

$$\|f\|_2 \leq k \sup_{t \in [0, 1]} |g(t)|.$$

Hint: Use the linear map induced by the left hand side of the equation.

5. Let $C^1[0, 1]$ be the Banach space of all continuously differentiable functions on $[0, 1]$ with norm

$$\|f\|_1 = \sup_{t \in [0, 1]} |f(t)| + \sup_{t \in [0, 1]} |f'(t)|, \quad f \in C^1[0, 1].$$

For fixed $g \in C^1[0, 1]$, consider the linear map $M_g : C^1[0, 1] \mapsto C^1[0, 1]$ given by $M_g f = gf$. Show that M_g is bounded and determine its spectrum.

6. Let $g \in C[0, 1]$ be fixed and let $V_g : C([0, 1]) \rightarrow C([0, 1])$ be defined by

$$V_g f(x) = \int_0^x g(t) f(t) dt.$$

Show that V_g is compact and determine its spectrum.